



How to use Sigma

Welcome, explorer! This guide will help you use Sigma to navigate the vast realms of Qeepsake's data warehouse with ease.

[Qeepsake's Sigma Homepage](#) [company link, redacted]

- For written learners, there's a detailed walkthrough of the basics of Sigma below.
- For visual learners, I've recorded some video walkthroughs where I go through and explain all of our Sigma folders, and build datasets, charts, and dashboards

 [Sigma video walkthroughs](#) [company link, redacted]

- For those who learn by doing, feel free to login to Sigma as a creator and explore! You can use Philip's account which is in the shared 1password. Please make sure to create a personal folder and save your work in there while you're learning instead of in the team folders.

[You can also check out Sigma's help center here.](#) I've written this guide because I think it's easier to learn Sigma for our purposes with Qeepsake data instead of the sample data they provide in [this tutorial](#), but there are tons more resources in the help center than I can provide here. Definitely bookmark it and use it as a reference when you're getting stuck trying to build something!

What is Sigma?

- Qeepsake's business intelligence tool. It's meant to be a central place for information about the business that *all teams* can use.
- A collection of **datasets (tables)** and **workbooks (dashboards)** that store important information about the business and display it in ways that are easy to visualize, monitor, and analyze

- A tool for working with tables in our database like you would work with a spreadsheet! With Sigma you can:
 - See any table in our database as if it were a spreadsheet (you can't edit the cells to change the data in our database, this is a setting that we have turned off)

Search

onboards

ENDORS

Overview

Columns

Links

Lineage

Worksheet

Materialization

Permissions

fx

BASE COLUMNS

123

User Id

Day Onboarded

abc Referral Code

abc Onboarding Channel

abc Oauth Id

abc Utm Campaign

abc Utm Term

abc Utm Source

abc Utm Content

abc App Version

766352

2021-12-14 00:00:00

OFDB

paid_social

null

lc_acq_saved_roas

acq_int_abc_motherhood

facebook

VID_P21_LookFamiliar

null

766034

2021-12-14 00:00:00

IntOrg

intentional_organic

null

social

instagram

bio_button

2.17.2-236978

766318

2021-12-14 00:00:00

OFDB

paid_social

null

lc_acq_saved_roas

acq_broad

facebook

VID_P20_BrandStory

null

766350

2021-12-14 00:00:00

OFDF

mobile

null

2.17.1-105288

766425

2021-12-14 00:00:00

OFDF

mobile

null

2.17.2-236978

766387

2021-12-14 00:00:00

OFDB

paid_social

null

lc_acq_saved_roas

acq_int_abc_motherhood

facebook

VID_P20_BrandStory

2.17.2-236978

766422

2021-12-14 00:00:00

OFDF

mobile

null

2.17.1-105288

766440

2021-12-14 00:00:00

IntOrg

intentional_organic

null

social

instagram

bio_button

2.17.2-236978

766309

2021-12-14 00:00:00

OFDF

mobile

null

2.17.2-236978

766372

2021-12-14 00:00:00

OFDB

paid_social

null

lc_acq_saved_roas

acq_int_abc_motherhood

facebook

VID_P20_BrandStory

2.17.2-236978

766324

2021-12-14 00:00:00

OFDB

paid_social

null

lc_acq_saved_roas

acq_broad

facebook

VID_P24_MostMoms_UGC

null

766336

2021-12-14 00:00:00

OFDB

paid_social

null

lc_acq_reg

acq_broad_reg

facebook

VID_P4_FAQ

2.17.2-236978

766340

2021-12-14 00:00:00

OFDA

organic

null

null

null

null

null

2.17.2-236978

766358

2021-12-14 00:00:00

OFDF

mobile

null

2.17.2-236978

766363

2021-12-14 00:00:00

OFDB

paid_social

null

lc_acq_reg

acq_master_lal1_reg

facebook

VID_P24_MostMoms_UGC

null

766181

2021-12-14 00:00:00

OFDB

paid_social

null

lc_acq_saved_roas

acq_int_abc_motherhood

facebook

VID_P20_BrandStory

2.17.2-236978

766487

2021-12-14 00:00:00

OFDF

mobile

null

2.17.2-236978

766499

2021-12-14 00:00:00

OFDA

organic

null

null

null

null

null

2.17.2-236978

766407

2021-12-14 00:00:00

OFDB

paid_social

10159348659006049

lc_acq_saved_roas

acq_int_abc_motherhood

facebook

VID_P22_Intro_SharkTank

null

766449

2021-12-14 00:00:00

OFDF

mobile

null

2.17.2-236978

766455

2021-12-14 00:00:00

OFDB

paid_social

null

lc_acq_saved_roas

acq_int_abc_motherhood

facebook

VID_P20_BrandStory

null

766466

2021-12-14 00:00:00

OFDA

organic

null

2.17.1-105288

766412

2021-12-14 00:00:00

OFDB

paid_social

null

LC_ACQ_LALS_ROAS

acq_fb_fbengagement_lal1

facebook

VID_P41_5star

null

766418

2021-12-14 00:00:00

OFDB

paid_social

null

lc_acq_saved_roas

acq_int_abc_motherhood

facebook

VID_P20_BrandStory

null

- Write formulas to calculate *new* metrics based off of different columns in your table

fx Concat([Utm Term], [Utm Source], [Utm Campaign])									
FILTERS		BASE COLUMNS							
No filters		123 User Id	Day Onboarded	abc Referral Code	abc Onboarding Ch...	abc Oauth Id	abc UTM String	abc	
PARAMETERS	No parameters	766352	2021-12-14 00:00:00	OFDB	paid_social	null	acq_int_abc_motherhoodface	lc_s	
		766034	2021-12-14 00:00:00	IntOrg	intentional_organic	null	instagramsocial	soc	
		766318	2021-12-14 00:00:00	OFDB	paid_social	null	acq_broadfacebooklc_acq_s	lc_s	
		766350	2021-12-14 00:00:00	OFDF	mobile	null			
		766425	2021-12-14 00:00:00	OFDF	mobile	null			

- Summarize data over different dimensions (like counting users onboarded by date or journals by user ID)

[onboards/day_onboarded]				
FILTERS ⓘ +		BASE COLUMNS		
filters		123 User Id ▾	📅 Day Onboarded ▾	abc Referral Code ▾
		766352	2021-12-14 00:00:00	paid_soc
PARAMETERS ⓘ +		766034	2021-12-14 00:00:00	intention
		766318	2021-12-14 00:00:00	paid_soc
parameters		766350	2021-12-14 00:00:00	mobile
		766425	2021-12-14 00:00:00	mobile
		766387	2021-12-14 00:00:00	paid_soc
		766422	2021-12-14 00:00:00	mobile
		766440	2021-12-14 00:00:00	intention
		766309	2021-12-14 00:00:00	mobile
		766372	2021-12-14 00:00:00	paid_soc
		766324	2021-12-14 00:00:00	paid_soc
		766336	2021-12-14 00:00:00	paid_soc
		766340	2021-12-14 00:00:00	organic
		766358	2021-12-14 00:00:00	mobile
		766363	2021-12-14 00:00:00	paid_soc
		766181	2021-12-14 00:00:00	paid_soc
		766487	2021-12-14 00:00:00	mobile
		766499	2021-12-14 00:00:00	organic
		766407	2021-12-14 00:00:00	paid_soc
		766449	2021-12-14 00:00:00	mobile
		766455	2021-12-14 00:00:00	paid_soc

fx CountDistinct([User Id])				
FILTERS ⓘ +		LEVEL 1 ⋮	BASE COLUMNS	
No filters		📅 Date ▾	123 Users Onboarded ▾	123 User Id ▾
		2021-12-14 00:00:00	181	abc Referral Code ▾
PARAMETERS ⓘ +				abc Onboarding Ch...
				766323 IntOrg intentional_organic
				766397 IntOrg intentional_organic
				766499 OFDA organic
				766448 OFDF mobile
				766494 OFDF mobile
				766181 OFDB paid_social
				766403 OFDF mobile
				766369 OFDF mobile
				766329 OFDB paid_social
				766381 OFDF mobile
				766383 OFDF mobile
				766371 OFDD google
				766492 OFDD google

- Link different sources of data together in an intuitive way (imagine using "vlookup" but way easier)

onboards

User Id ▾	User Id (Conversions) ▾	Day Onboarded ▾	Referral Code ▾	Onboarding Channel ▾	
766352	null	2021-12-14 00:00:00	OFDB	paid_social	null
766034					
766318					
766350					
766425					
766387					
766422					
766440					
766309					
766372					
766324					
766336					
766340					
766358					
766363					
766181					
766487					
766499					
766407					
766449					
766455					

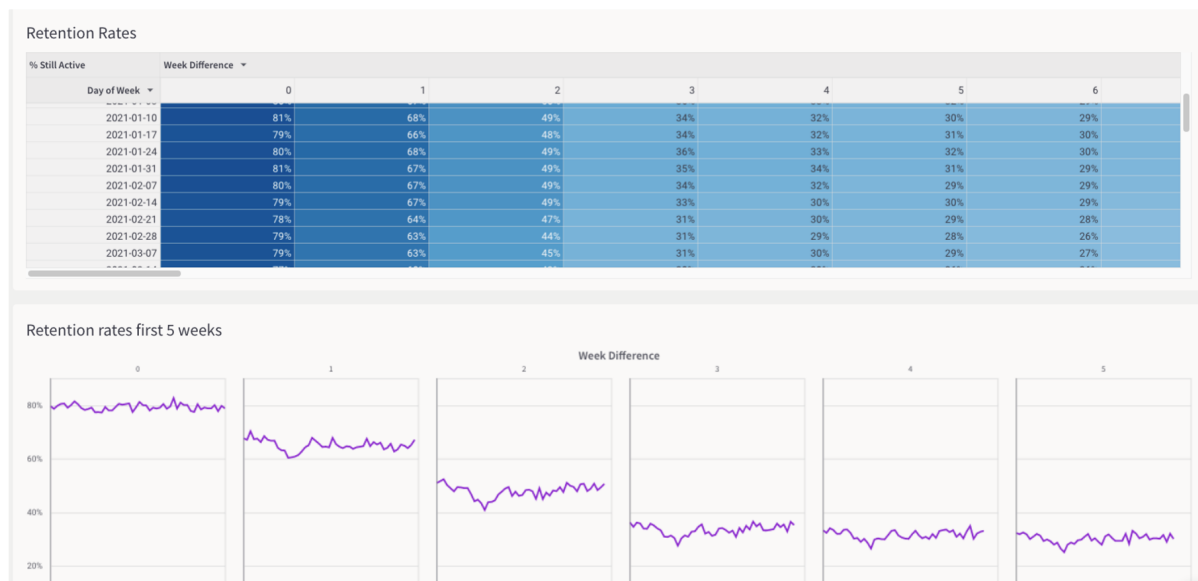
LINKED RECORD

conversions

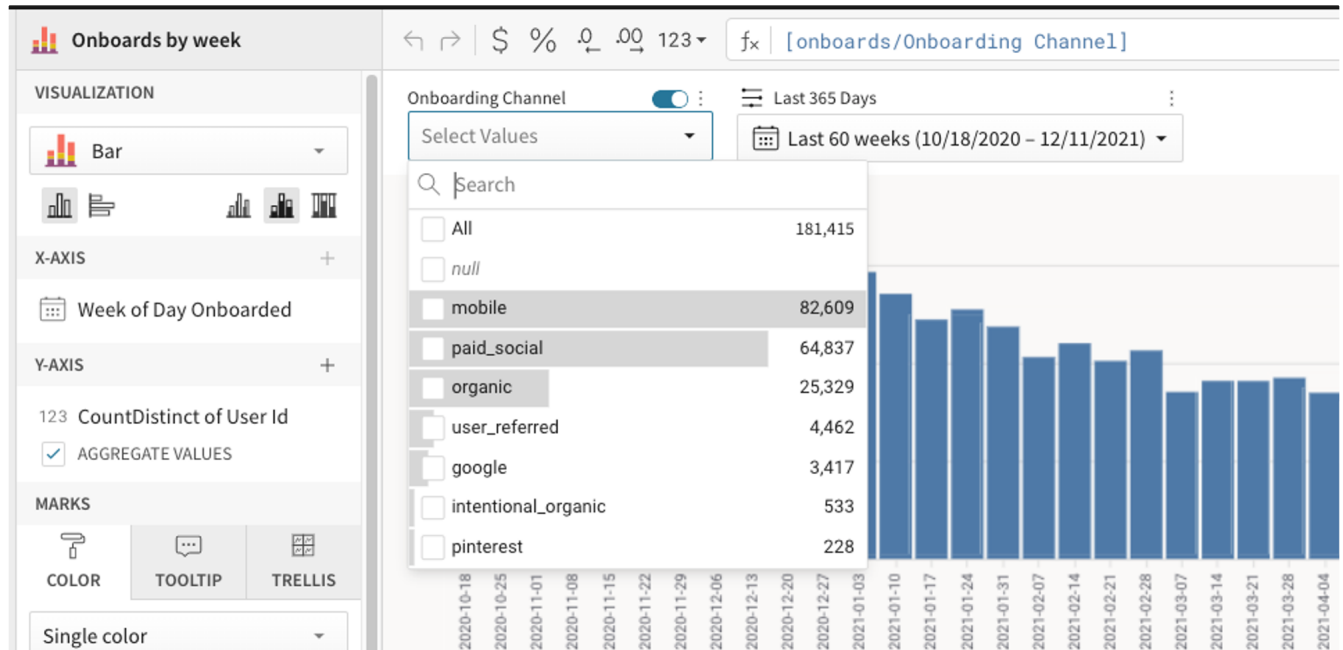
766034

- ☒ User Id 766034
- ☐ Subscription Name Qeepsake Plus
- ☐ Payment Provider ios_iap
- ☐ Day Subscription Starts 2021-12-14 00:00:00
- ☐ Is First Paid Sub 1
- ☐ Day of Conversion 0
- ☐ Discount Amount 0
- ☐ Revenue Transacted 47.88
- ☐ Day Onboarded 2021-12-14 00:00:00
- ☐ Referral Code IntOrg

- Create charts based off of your tables



- Filter any column in a table or a chart to drill down into a specific subset of your data.



How do I get started?

- Sign up for a "viewer" account by asking one of the administrators (Ryan or Philip) to invite you via email.
- Check out the "Workspaces" folder and the "All Endorsed Datasets" folder within it
 - **Workspaces** is where all team-shared data should live. There are folders for each business team, plus some additional folders like Financial modeling, All Endorsed Datasets, and Teamwide.
 - **All Endorsed Datasets** is essentially our "database" in Sigma. Data team curates this folder to make sure it contains the tables each team would need to do analysis and build dashboards, while maintaining a structure that makes sense and documenting it [here](#)
- Log into an "explorer" or "creator" account via the team's shared 1password (we have limited accounts here because we have to pay for them). Right now, the accounts we have are:
 - Creators (can create & edit folders, workbooks and datasets, can invite other users): Ryan, Philip
 - Explorers (can break down the data within workbooks and save different versions): Jeff, Duncan

- Try creating your own datasets and workbooks!
 - If you have a business analysis you need to do and it's important to do it quickly and correctly, set aside some time with Ryan to do a shared session where you'll build it together.
 - If not, you should still spend some time exploring our data and answering your own questions. **This is valuable time to the business — Being comfortable in Sigma will make you a more effective decision maker and a more informed team member!** What are you curious about? Here are some questions you could try to answer:
 - | How many users have we added per year of Qeepsake's existence?
 - | What is the average number of journals a Qeepsake user has?
 - | What's the most common journal name?
 - | How many questions has the average user gotten over their lifetime? How many have they answered?
 - | What is the average price of a subscription counting discounts from sales vs. not counting them?
 - | What % of users answer a question in their Xth (1st, 6th, 18th, etc) week on Qeepsake?
 - | How many questions are answered with a response that contains the word "love"?
 - | How many times does the average user tap the "new entry" button in the app during their first week?
 - | Who has more journals on average - People whose youngest journal is less than a year old, or people whose youngest journal is 1-2 years old?

Okay, I'm poking around, but still don't really know what I'm supposed to be doing. Can you help me out?

Sure thing!



First thing, you might want to watch some of the walkthrough videos:

 [Sigma video walkthroughs](#) [company link, redacted]

But if you're more of a written learner, let's go through some of the basic functions in Sigma and how to do them. Let's pretend we're answering one of the questions above:

"How many questions has the average user gotten over their lifetime? How many have they answered?"

And we're starting on our creator account's homepage. The first thing we're going to do is go to "my documents" and create a folder to store our analysis:

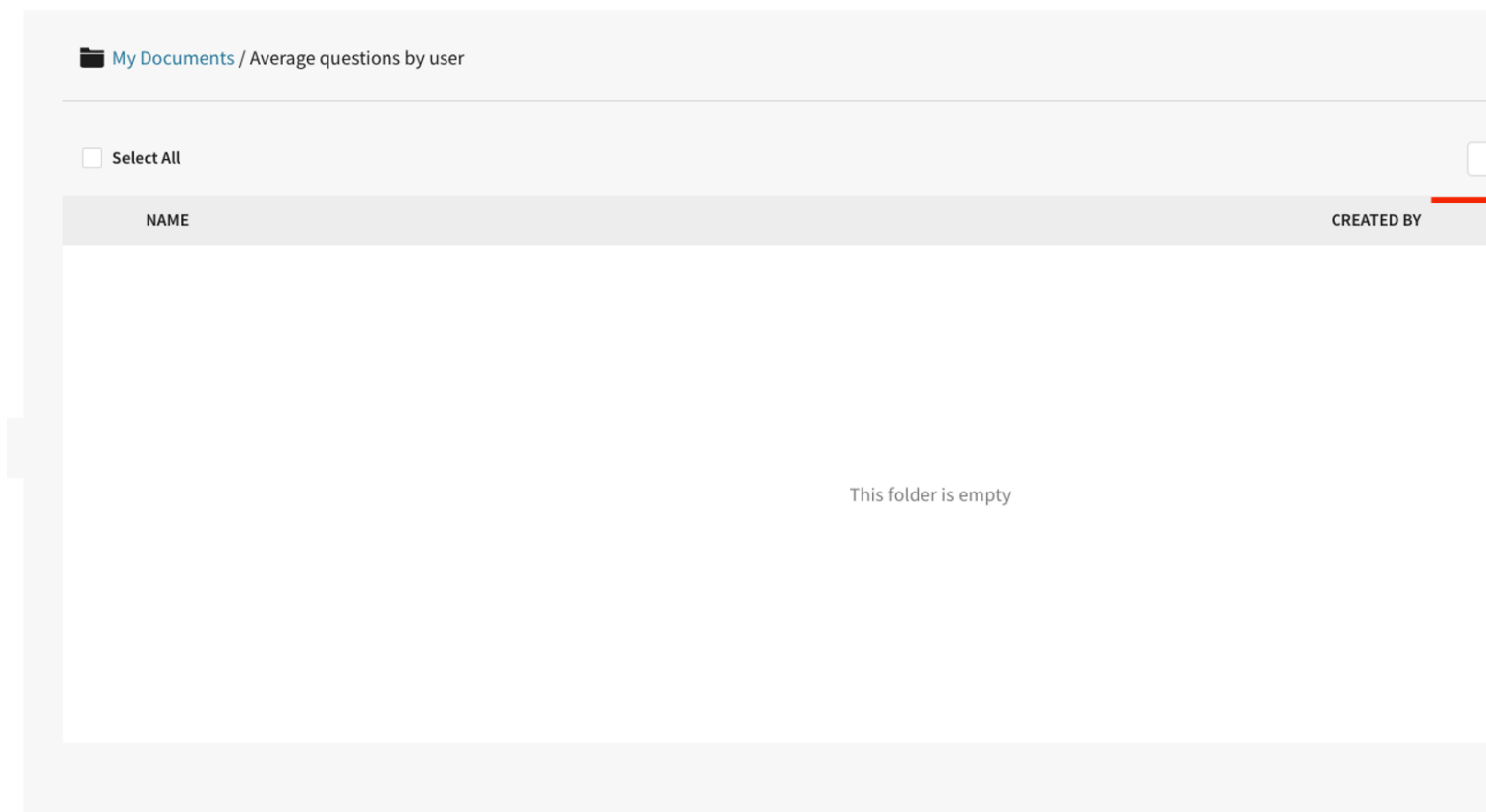
The screenshot shows the Sigma dashboard homepage. The left sidebar contains a 'Create New' button, a 'Home' link, and a list of navigation items: 'Recents', 'Favorites', 'Shared with Me', 'Templates', 'My Documents' (highlighted with a red arrow), 'Workspaces', and 'CONNECTIONS' (which includes 'Bandwidth SMS App' and 'BigQuery'). The main content area is titled 'RECENTS' and displays a table of recent documents. The table has columns for 'NAME' and 'MY LAST ACTIVITY'. The documents listed include 'Weekly KPI', 'onboards', 'Book Sales Reporting', 'Stripe Charges Without Corresponding Book Orders', 'book_orders', 'book_orders_with_analysis_info', 'book_eligibility', and 'book_vouchers'. Below the table, there is a section titled 'EXPLORE EXISTING DOCUMENTS' with three document cards: 'All Channels - Subscription Targets', 'Recap Recipient Button Usage Analysis', and 'Exploration'.

NAME	MY LAST ACTIVITY
Weekly KPI	Viewed 16 hours ago
onboards	Used 17 hours ago
Book Sales Reporting	Updated 17 hours ago
Stripe Charges Without Corresponding Book Orders	Used 17 hours ago
book_orders	Used 17 hours ago
book_orders_with_analysis_info	Updated 17 hours ago
book_eligibility	Updated 17 hours ago
book_vouchers	Updated 17 hours ago

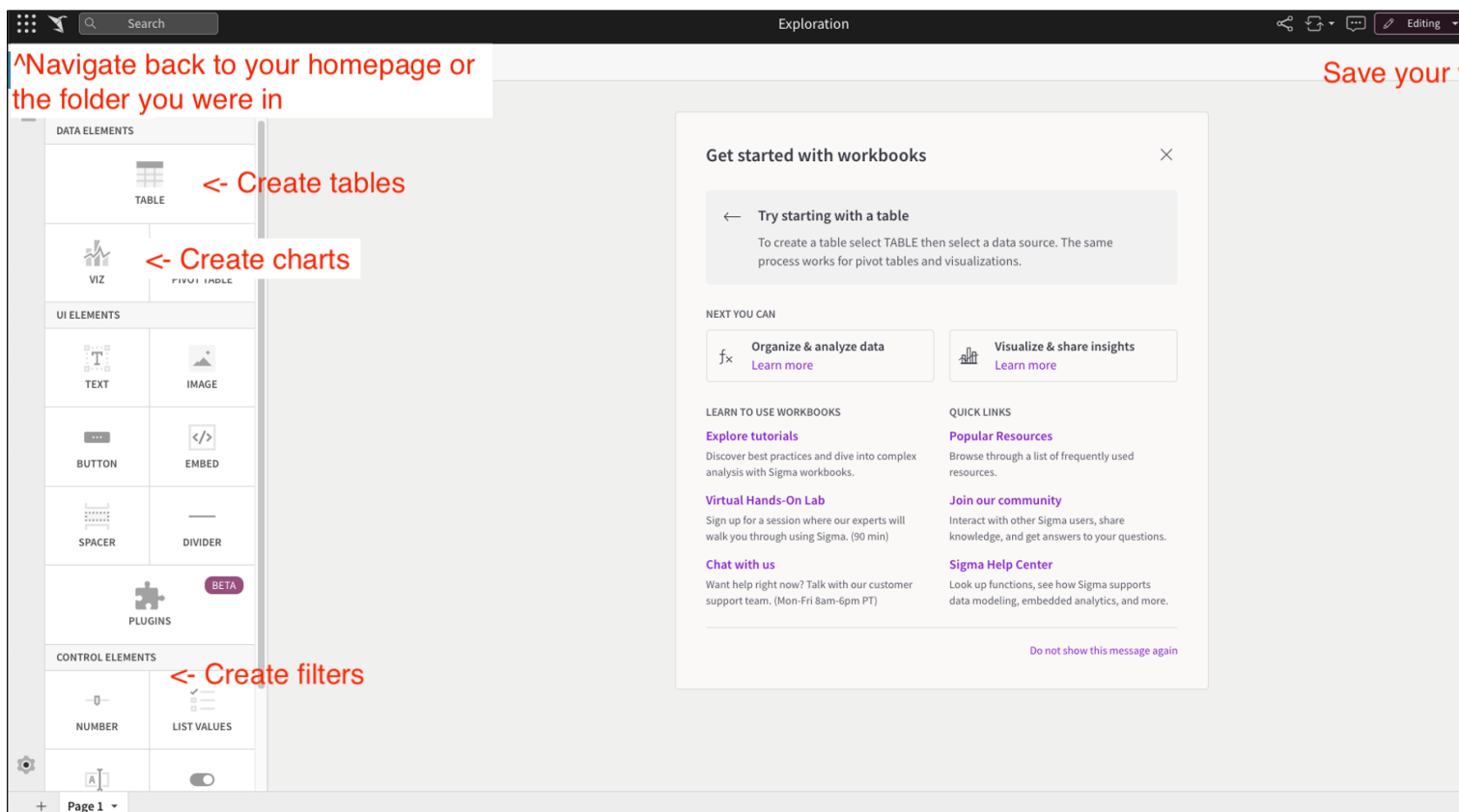
The screenshot shows the 'My Documents' page in Sigma. The left sidebar is identical to the previous screenshot, with 'My Documents' highlighted. The main content area is titled 'My Documents' and features a 'Select All' checkbox. To the right of the 'Select All' checkbox is a red arrow pointing to a 'Create New' button. Below this, there is a table with columns for 'NAME' and 'CREATED BY'. The table contains one document: 'New Proposed Qeepsake Questions - Recommended Edits.csv' created by 'Ryan Condon'.

NAME	CREATED BY
New Proposed Qeepsake Questions - Recommended Edits.csv	Ryan Condon

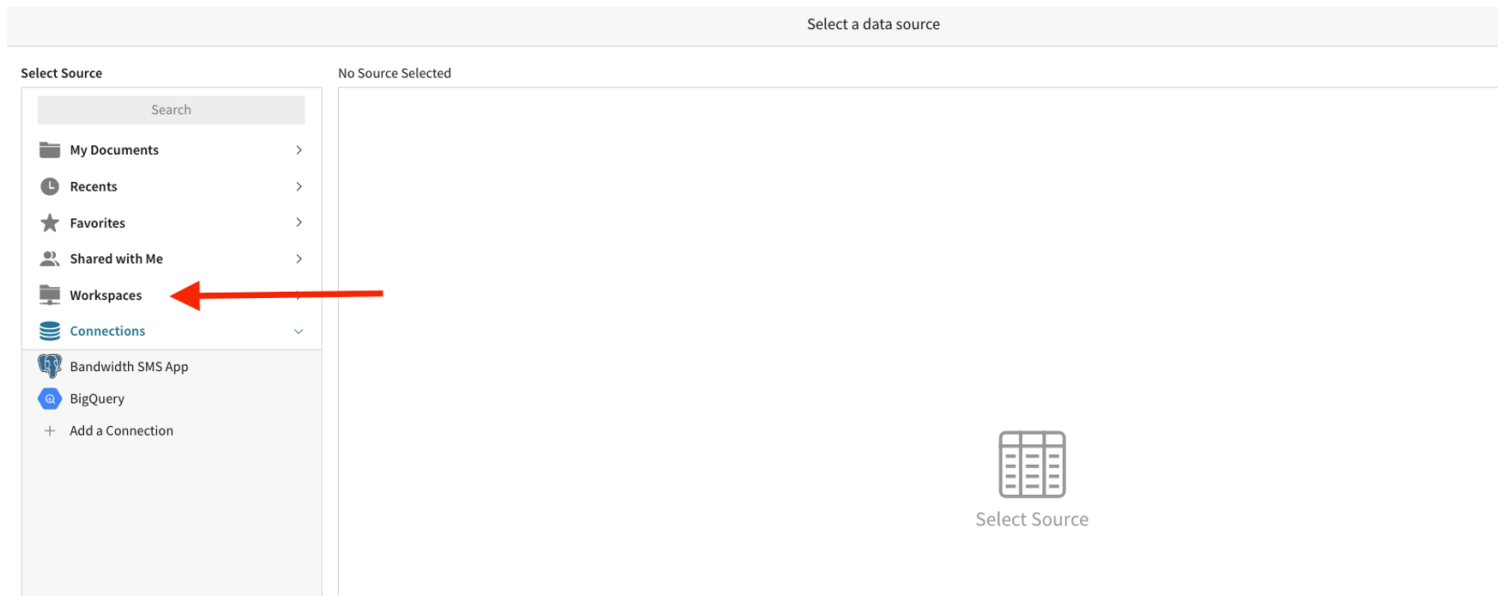
Then we'll create a new workbook in this folder. Workbooks (AKA dashboards) are generally where your charts and final analysis will live. If we need to create a custom table to build our analysis off of, we'd create a dataset in this same folder for that purpose.



This is the workbook opening screen. I've labeled a few key places in the UI where most of the important things happen:



Usually it's good to start by creating a table. Let's bring in the journal entries table. Sigma gives you an option to select from your existing tables, write SQL to pull directly from our database, or upload a CSV. This table already exists (technically it's saved as a dataset in Sigma, we'll use the terms "table" and "dataset" pretty interchangeably), so we select "Tables and datasets". That opens up this screen:



You will **almost always** want to navigate to **Workspaces→Analytics database** so that you're working with existing data that the data team has already curated. The journal entries table is in the "Engagement" folder, but if I didn't know that I could look it up here (even if I was looking for something as vague as "entries", I could control-F and search the page)

Selecting the table lets us preview it before we decide it's the one we want:

22 of 22 selected

Select a data source

Select Source

Search

< Back Engagement

- engagement_curves_table
- journal_entries_table**
- lifetime_entries
- Retention Rates
- user_response_history
- users_with_entries_last_month

Select Columns

Search...

☒ Select All 22 of 22 selected

- ☒ 123 Journal Entry Id
- ☒ Day Created
- ☒ 123 User Id
- ☒ 123 Question Id
- ☒ abc Question
- ☒ abc Response
- ☒ 123 Response Length
- ☒ abc Source
- ☒ Meaningful
- ☒ Unprompted
- ☒ With Attachment
- ☒ 123 Journal Id
- ☒ Journal Created At
- ☒ Journal Birthdate
- ☒ 123 Journal Age When Entry ...
- ☒ abc Onboarding Channel
- ☒ abc App Version
- ☒ abc Active Subscription
- ☒ Milestone
- ☒ Milestone (1)

journal_entries_table

123	Journal Entry Id	Day Created	123	User Id	123	Question Id	abc	Question	abc	Response	123
	34478990	2020-11-25 00:00:00		580620		736	How were the first weeks of t	Nursing Samuel was a challer			
	34501934	2020-11-25 00:00:00		566789		709	What special time did you ha	Well, Charli slept for 7 hours t			
	34505745	2020-11-25 00:00:00		15361		966	Where are you planning on ha	We are doing thanksgiving at			
	34486037	2020-11-25 00:00:00		118109		965	What is %yc's favorite Thanks	Sweet potato			
	34501463	2020-11-25 00:00:00		191955		966	Where are you planning on ha	Staying at home because of C			
	34497329	2020-11-25 00:00:00		479676		964	Are you and %yc celebrating	My mom, me and Arianna. Sn			
	34472666	2020-11-25 00:00:00		398575		964	Are you and %yc celebrating	Yes, in Raleigh with mom and			
	34492994	2020-11-25 00:00:00		533104		991	How do you think your spous	He was a great support durin			
	34495146	2020-11-25 00:00:00		427765		924	Does %s have a best friend?	Louie Marco and Lachlan!			
	34471364	2020-11-25 00:00:00		493153		659	What was one thing you enjo	Helping her go through her cl			
	34498028	2020-11-25 00:00:00		513255		966	Where are you planning on ha	At home.			
	34492693	2020-11-25 00:00:00		24955		964	Are you and %yc celebrating	It will just be us, Babi and Dzi			
	34493440	2020-11-25 00:00:00		509457		1124	Have you snapped a sweet pl				
	34492418	2020-11-25 00:00:00		92967		964	Are you and %yc celebrating	Mason, his daddy, and I will b			
	34491786	2020-11-25 00:00:00		221359		964	Are you and %yc celebrating	We are celebrating with Jeffe			
	34490053	2020-11-25 00:00:00		109221		964	Are you and %yc celebrating	Sort of. We aren't allowed to r			
	34475579	2020-11-25 00:00:00		555259		813	Is there a song you like to pla	Baby shark, over and over agi			
	34488161	2020-11-25 00:00:00		185579		965	What is %yc's favorite Thanks	Hmm. Pasta... haha			
	34489151	2020-11-25 00:00:00		551270		785	Has %s learned a new skill? If	How to say "no."			
	34491473	2020-11-25 00:00:00		490652				Helping make cupcakes for M			
	34502031	2020-11-25 00:00:00		421520				Sweet snuggles			
	34490954	2020-11-25 00:00:00		406353		847	What was something cute %s	She has been giving kisses to			
	34492595	2020-11-25 00:00:00		457852		87	What is something %s is afrai	Sometimes she is afraid of bi			
	34486877	2020-11-25 00:00:00		16387		965	What is %yc's favorite Thanks	Sweet potato casserole with i			
	34484802	2020-11-25 00:00:00		583474		295	What is %s's favorite subject	History			
	34474160	2020-11-25 00:00:00		370645		964	Are you and %yc celebrating	yes in our new house ❤️			
	34498710	2020-11-25 00:00:00		473345		369	Has %s invented any special	Riding the horsey on your leg			
	34498530	2020-11-25 00:00:00		557759		1279	Describe a recent snuggle or	When he sleeps on my chest			
	34489162	2020-11-25 00:00:00		576680		733	What was the best part of yo	Peyton's been very sleepy late			
	34496684	2020-11-25 00:00:00		86256		964	Are you and %yc celebrating	We are going to celebrate at			
	34485832	2020-11-25 00:00:00		92363		965	What is %yc's favorite Thanks	She loves Turkey, dressing, cc			
	34473943	2020-11-25 00:00:00		406717		391	What is the funniest thing %s	Today he walked around tapp			
	34505567	2020-11-25 00:00:00		581347		966	Where are you planning on ha	As with most things in 2020,			
	34487174	2020-11-25 00:00:00		512787		329	Who do you hope will be one	Her aunt Laurie and aunt Mar			
	34484964	2020-11-25 00:00:00		582363		258	What are some names/nickn	Linebacker baby, boo boo, ill f			
	34500571	2020-11-25 00:00:00		456556		675	Does %s have a lovey? Descri	She doesn't really have a love			
	34478645	2020-11-25 00:00:00		37223		964	Are you and %yc celebrating	Josh's side of the family			

The first column in every table should (almost always) be some sort of ID column that represents the basic item that each row in this table represents. In this case, each row of this table is a unique journal entry created by a user. We'll be able to summarize this data by users, since we see that there's also a column identifying which user ID created each entry. So we click "done" and see this table back in our workbook:

journal_entries_table

GROUPINGS 4

COLUMNS

1

2

3

5

6

Journal Entry Id

Day Created

User Id

Question Id

Question

Response

Response Length

Source

Meaningful

Unprompted

SUMMARY ^ 46,248,373 Rows - 22 Columns

Some key areas to notice:

1. You can view the columns in the table either in the table itself or on the left while you have the table selected.
2. You can view the underlying dataset that your table is pulling from in the bottom left. **The new table you created in your workbook is not the same as the dataset! It is a "child object" of it. What that means is: any edits you make to your new table will *not* be applied to the dataset itself, but edits to the dataset will apply to the table. For example, you can choose to only bring in certain columns from the dataset into your table, and any formatting you do on those columns will not affect the original columns in the dataset.**
3. You can select/edit any individual column by clicking on its dropdown.
4. You can still add new objects to the workbook with the "+" button in the top left.
5. You can write custom formulas to create new columns (or edit existing ones) in the formula bar up top
6. Just like in Excel, there's a formatting bar as well with buttons for common formatting options.

Let's create a new column in our table with a custom formula. Remember the question we're trying to answer is

"How many questions has the average user gotten over their lifetime? How many have they answered?"

So let's create a column to see if the user answered the question. To create a new column, we click the "+" button next to columns in the top right of the bar on the left where all the columns are listed, and select "new column". It will show up at the end of our table as "Calc" (we can rename it by double-clicking on the column name). We can write a formula for this column. We'll count it as a response if a user sends in a text response, or even if they only send in a photo. Something isn't a response if and only if both of those columns are "Null".

Here is the function index for Sigma - If you want to do a certain kind of calculation on a column but aren't sure how, start here!

Our function will be

```
isnull(response) and withattachment = false
```

This will return "true" if the response column is null and the withattachment column is false.

The screenshot shows a data exploration tool interface. At the top, a formula bar contains the expression: `IsNull([Response]) and [With Attachment] = False`. A red arrow points to this formula. Below the formula bar, a data table titled "journal_entries_table" is displayed. The table has columns: Journal Birthdate, Journal Age When Entr..., Onboarding Channel, App Version, Active Subscription, Milestone, Milestone (1), and Question Last Updated. The table contains multiple rows of data. On the left side, a sidebar shows a list of columns and a "Calc" button, which is highlighted with a red arrow.

Let's look at what we can do with our new column by selecting the dropdown menu (I renamed it "Responded?" but it should actually be "Didn't Respond?" because a true value indicates a question that WASN'T responded to)

The screenshot shows a dropdown menu for a column named "Responded?". The menu options are: SORT, Filter, Add new column, Add column via lookup... (BETA), Duplicate column, Rename column, Hide column, Delete column, Group column, Column details..., Transform, and Conditional formatting. Red numbers 1 through 7 are placed next to the first seven options respectively.

1. We can sort the column in ascending or descending order (this will sort the entire table by this column)
2. We can filter the table by the values in this column
3. We can always add a new column via this dropdown menu
4. We can duplicate the column (especially useful if you have written a long, complicated formula that you want to iterate on but don't want to have to write all over again)
5. We can hide or delete the column (hiding just removes it from visibility in the table in the workbook, deleting it removes it from the table altogether *(but not from the underlying dataset!)*)

6. We can group the entire table by this column. More on this soon!
7. We can examine the values in the column by looking at "column details". Let's do that so we can see what our formula is showing!

Column Details			✕
About		Summary	
📄 Responded?		True	0.0% 0
✖ IsNull([Response]) and [With Attachment] = False		False	100.0% 46,248,373
		Nulls	0.0% 0
		Row Count	100.0% 46,248,373
			Learn more about Columns

It looks like there isn't a single "true" value in the entire 46-million row table. That's crazy! Well, it turns out that we've already filtered out non-responses from this table since we don't count them as "journal entries". You can see this in the [table documentation](#):

Journal Entries

This is a cleaned-up version of the `journal_entries` table in our [Heroku database](#) that to only non-null entries from users who are currently on one of our standard plan types. It also includes additional calculated fields that flag useful properties of the journal entry.

- Journal entry ID (used for count distincts)
- User ID
- Question ID (null = Unprompted)
- Day the entry was created
- Rendered text (This column is buggy but will sometimes display the text of the question asked)
- Response (The user's response)

It's always important to remember that in Sigma, you're working with curated data, not a direct representation of our database. Always check the documentation if you're seeing something you don't expect! And feel free to reach out to a data team member if you have questions about why data is filtered in a certain way, or whether it can be changed.

So it looks like we won't be able to answer the first part of our question,

"How many questions has the average user gotten over their lifetime?"

with this table. However, we should be able to answer part 2, **How many have they answered?".** All we need is to find the total number of users, and the total number of journal entries lifetime, then divide the second by the first!

Whenever we want to find data that summarizes an entire table, what we'll want to do is GROUP the table by a column with only one value in it. That way, any metrics we calculate based off of that column's one value will be based off of the entire table! Luckily, we already have such a column. Let's group by it!

Search

journal_entries_table

GROUPINGS +

Responded? ←

CALCULATIONS +

COLUMNS +

123 Journal Entry Id

Day Created

123 User Id

123 Question Id

abc Question

abc Response

123 Response Length

abc Source

Meaningful

journal_entries_table

Responded?	Journal Entry Id	Day Created	User Id
- false	133244679	2021-05-03 00:00:00	6
	133192433	2021-05-03 00:00:00	6
	133190454	2021-05-03 00:00:00	1
	133133787	2021-05-03 00:00:00	6
	133152603	2021-05-03 00:00:00	6
	133103157	2021-05-03 00:00:00	3
	133162729	2021-05-03 00:00:00	6
	133171410	2021-05-03 00:00:00	4
	133210486	2021-05-03 00:00:00	6
	133220268	2021-05-03 00:00:00	1
	133194474	2021-05-03 00:00:00	6
	133177643	2021-05-03 00:00:00	6
	133213559	2021-05-03 00:00:00	6

Now we can see that all the way on the left of the table, we've taken this column and added it to a new "grouping level" in the table (separated by the gray divider line). Any columns we add at this level will be calculated as aggregations for each value in the "Responded?" column (in this case, there is only one value, "false", so any calculation will be over any row that has value "false", which is the entire table)

We can add new columns to calculate how many user IDs show up in this table, and how many journal entries there are. (You could also find the total journal entries by seeing how many rows are in the table, because each row corresponds to a single entry).

```
countdistinct(user id) countdistinct(journal entry id)
```


journal_entries_table

GROUPINGS

Responded?

CALCULATIONS

123 Total Users

123 Total Journal Entries

123 Avg entries per user

COLUMNS

123 Journal Entry Id

Day Created

123 User Id

journal_entries_table

Responded?	Total Users	Total Journal Entries	Avg entries per user	Journal Entry Id	Day Created	User Id	Question Id
false	606,578	45,923,666	76	34492418	2020-11-25 00:00:00	92967	964 A
				34488161	2020-11-25 00:00:00	185579	965 W
				34489151	2020-11-25 00:00:00	551270	785 H
				34491473	2020-11-25 00:00:00	490652	null ni
				34492595	2020-11-25 00:00:00	457852	87 W
				34486877	2020-11-25 00:00:00	16387	965 W
				34484802	2020-11-25 00:00:00	583474	295 W
				34498710	2020-11-25 00:00:00	473345	369 H
				34498530	2020-11-25 00:00:00	557759	1279 D

But let's try to answer the question in a different way, that might be slightly more useful. It would be great to see not just the average number of journal entries per user, but the *distribution*. To do this, we'll need to calculate the number of entries each user has in this table, and then graph each of those numbers. Luckily, this is incredibly easy to do in Sigma! All we have to do is GROUP again by user ID, and we can sum up the number of entries that each user has. So we select the dropdown on the user ID column, and group:

journal_entries_table

GROUPINGS

Responded?

CALCULATIONS

123 Total Users

123 Total Journal Entries

123 Avg entries per user

123 User Id

COLUMNS

123 Journal Entry Id

Day Created

123 Question Id

abc Question

journal_entries_table

Responded?	Total Users	Total Journal Entries	Avg entries per user	User Id	Journal Entry Id	Day Created	Question Id
false	606,578	45,923,666	76	61	4895849	2017-07-13 00:00:00	
					7521164	2017-12-23 00:00:00	
					735334	2016-06-30 00:00:00	
					22238587	2019-09-23 00:00:00	
					5449626	2017-08-19 00:00:00	
					13642878	2018-10-02 00:00:00	
					430284	2016-05-13 00:00:00	
					26983871	2020-04-08 00:00:00	
					26983832	2020-04-08 00:00:00	
					430345	2016-05-13 00:00:00	
					8491181	2018-02-13 00:00:00	
					2670884	2017-01-27 00:00:00	
					18235398	2019-04-10 00:00:00	
					6557495	2017-10-28 00:00:00	
					1530661	2016-09-28 00:00:00	

Sidenote: At this point I realized my "responded?" column should really say "true" instead of "false", so I added a not() surrounding the entire formula. The will change in the following screenshots!

Notice that the grouping by user ID doesn't remove the grouping by "Responded?" — It goes *under* it, as a subset of the original grouping. Since "Responded?" is grouping the entire table, that doesn't particularly matter here, but for example, if we added another grouping by "day created", what we would actually be grouping by is the combination of both *user* and *day*.

Next, we'll add a "journal entries for this user" column at our "user" grouping level.

journal_entries_table

GROUPINGS

Responded?

CALCULATIONS

123 Total Users

123 Total Journal Entries

123 Avg entries per user

123 User Id

CALCULATIONS

123 Journal entries for this user

COLUMNS

123 Journal Entry Id

Day Created

123 Question Id

abc Question

abc Response

123 Response Length

abc Source

Meaningful

Unprompted

With Attachment

123 Journal Id

Journal Birthdate

journal_entries_table

Responded?	Total Users	Total Journal Entries	Avg entries per user	User Id	Journal entries for this ...	Journal Entry Id
true	606,578	45,923,666	76	61	991	42
						15099
						43
						382
						43
						43
						43
						2221
						1548
						43
						877
						43
						43
						43
						43
						2698
						12
						43
						2698
						43
						1396
						73

SUMMARY ^ 46,248,373 Rows - 26 Columns

User ID 61 happens to be Jeff, and we can see that he has 991 lifetime entries. Impressive!

It would be nice if we didn't have to see all the journal entries associated with each user, and instead could just see a list of all our user IDs. Sigma lets us do that! We just have to collapse the lowest grouping level by clicking on the "-" sign next to user ID:

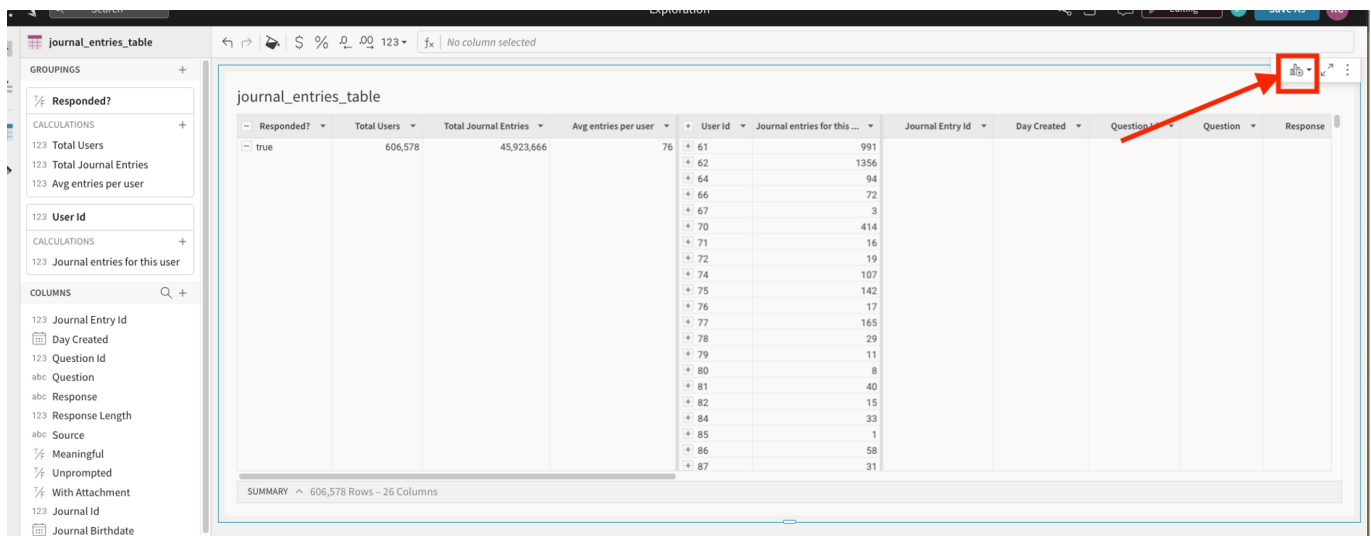
journal_entries_table

Responded?	Total Users	Total Journal Entries	Avg entries per user	User Id	Journal entries for this ...	Journal Entry Id	Day Created	Question Id	Question	Response
true	606,578	45,923,666	76	61	991					
				62	1356					
				64	94					
				66	72					
				67	3					
				70	414					
				71	16					
				72	19					
				74	107					
				75	142					
				76	17					
				77	165					
				78	29					
				79	11					
				80	8					
				81	40					
				82	15					
				84	33					
				85	1					
				86	58					
				87	31					

SUMMARY ^ 606,578 Rows - 26 Columns

Notice that you can still see the columns of the underlying table (and you can still do things like group by them or filter by them), but the underlying data has been wrapped up so that we can see things at the level of user IDs.

Awesome. Now let's graph it! To create a visualization based off of a table, we just click the "+chart" icon in the top right of the table.

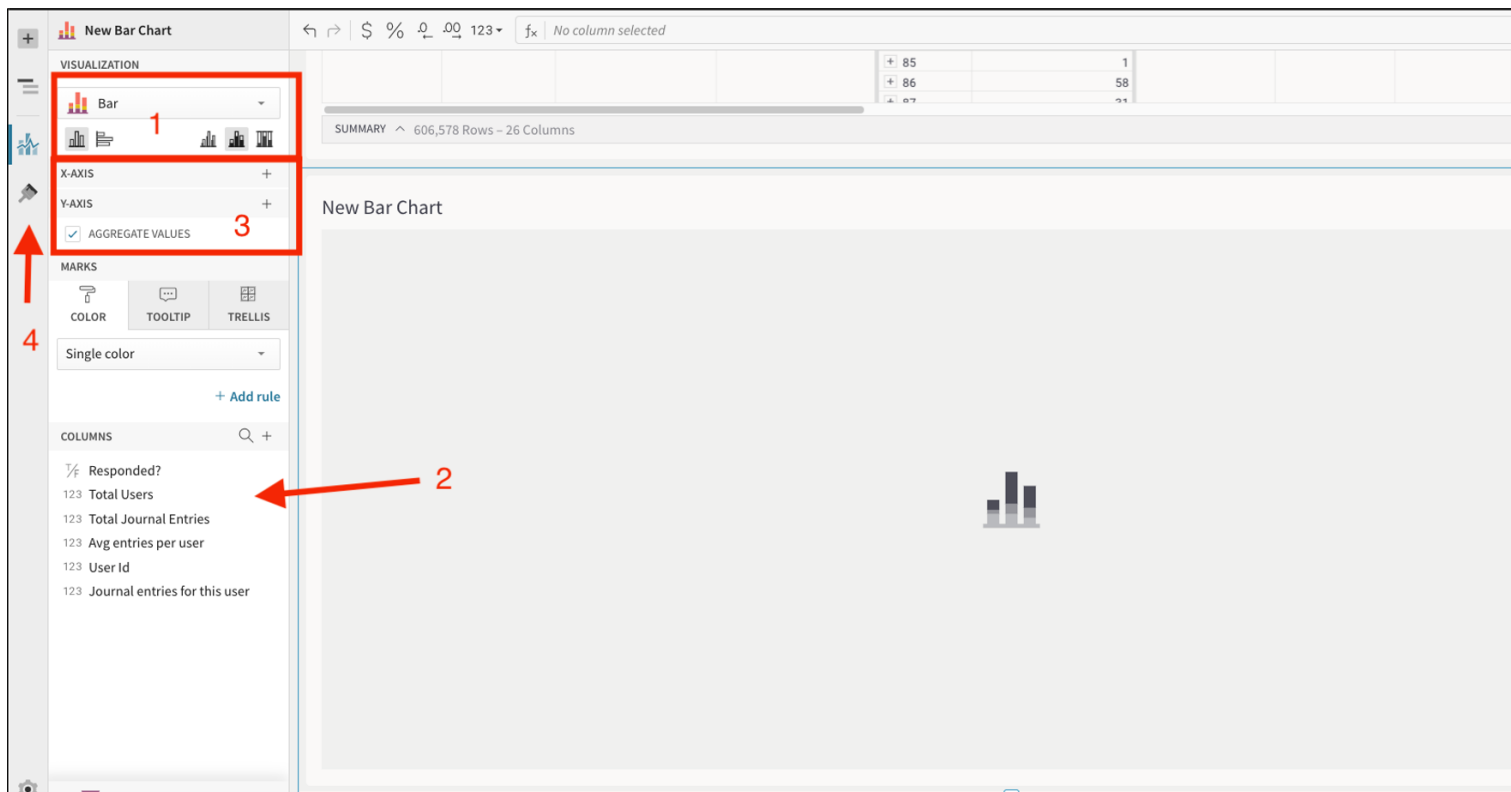


journal_entries_table

Responded?	Total Users	Total Journal Entries	Avg entries per user	User Id	Journal entries for this ...	Journal Entry Id	Day Created	Question	Response
+	606,578	45,923,666	76	+ 61	991				
				+ 62	1356				
				+ 64	94				
				+ 66	72				
				+ 67	3				
				+ 70	414				
				+ 71	16				
				+ 72	19				
				+ 74	107				
				+ 75	142				
				+ 76	17				
				+ 77	165				
				+ 78	29				
				+ 79	11				
				+ 80	8				
				+ 81	40				
				+ 82	15				
				+ 84	33				
				+ 85	1				
				+ 86	58				
				+ 87	31				

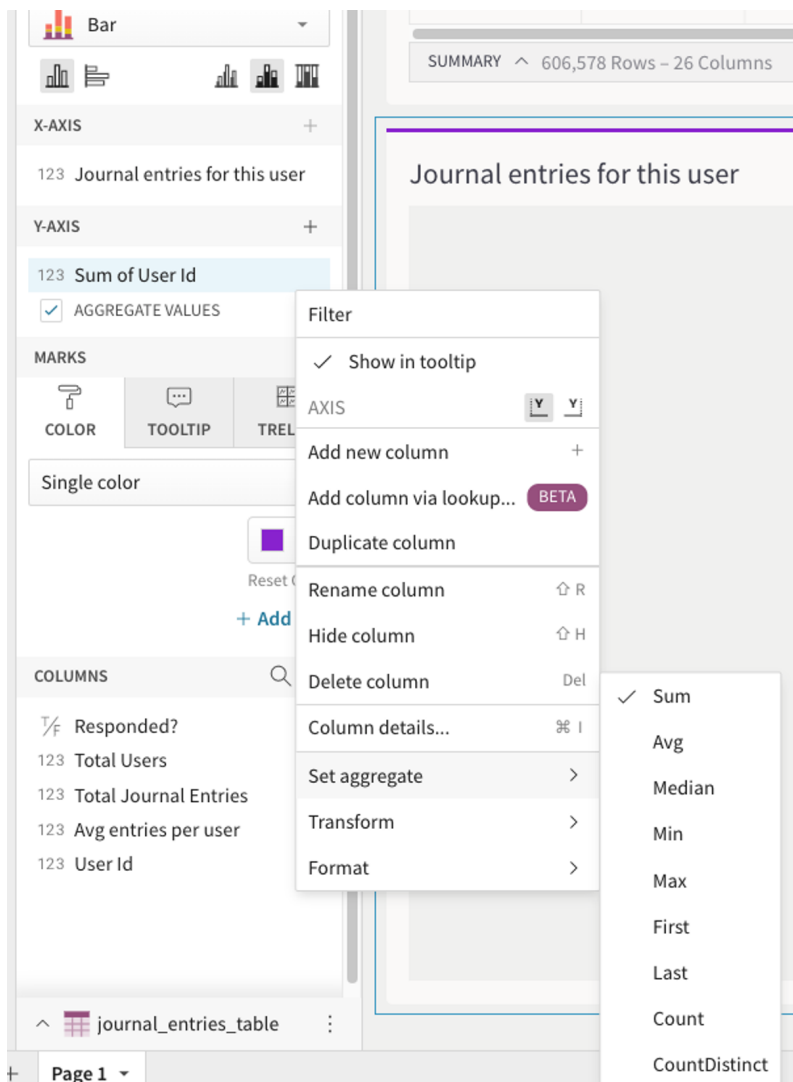
SUMMARY ^ 606,578 Rows - 26 Columns

This creates a "child element" of the table that will reflect the values in the table as the table gets changed, filtered, etc.

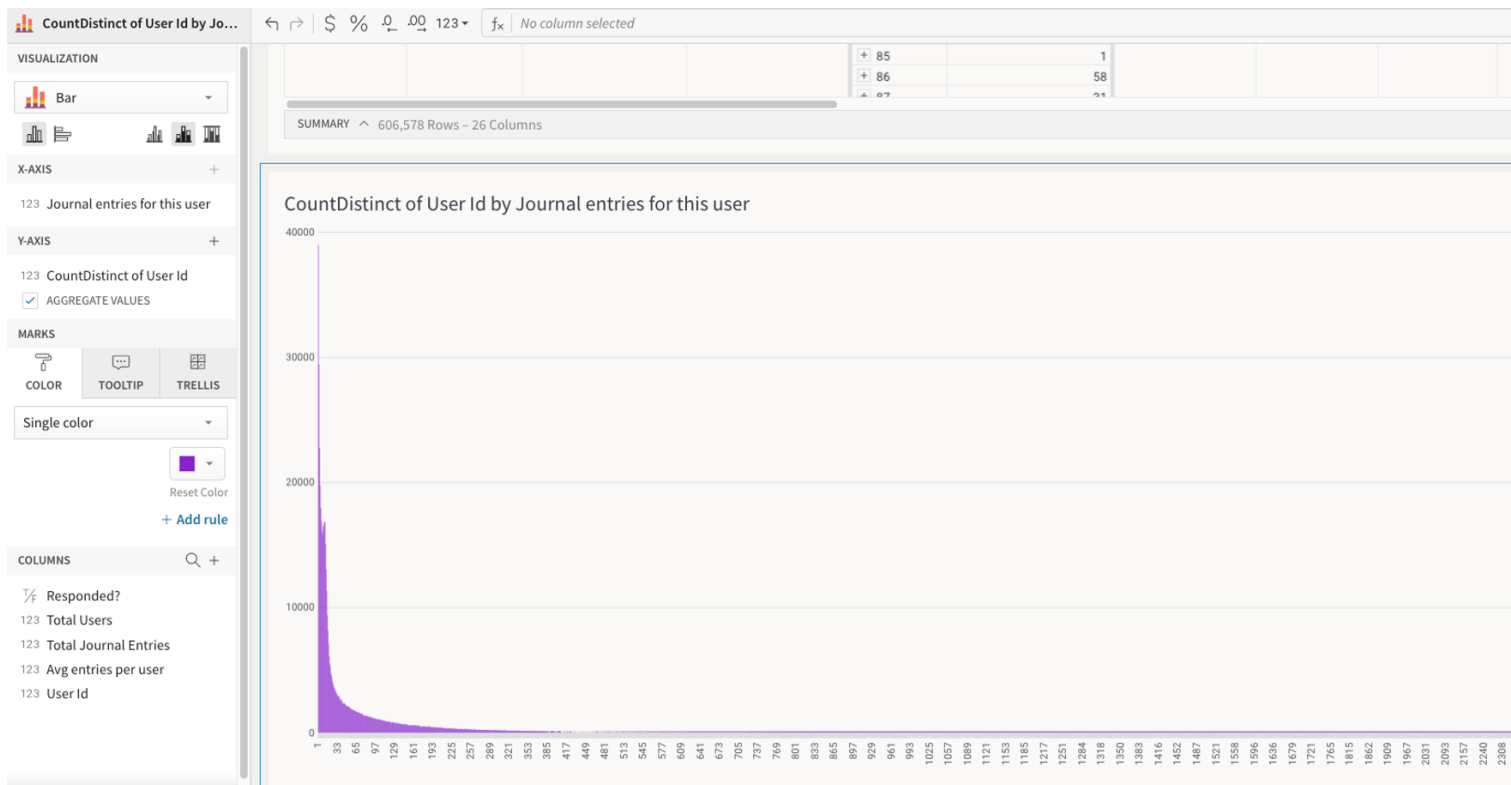


1. This is where you can choose which type of visualization you're creating. For our purposes, we're fine with a bar chart to show a distribution of integer values.
2. These are the columns from your original table. You can drag them up into...
3. The X and Y axes of your chart! You can drag existing columns into these areas to display them on the graph, but you can also write new columns using formulas.
4. The chart formatting section can be found by clicking on the paintbrush.

We want to see the distribution of user IDs based on how many entries they've made. So our X-axis will be "journal entries for this user", and our Y-axis will be counting the number of user IDs with that number of entries, like so:

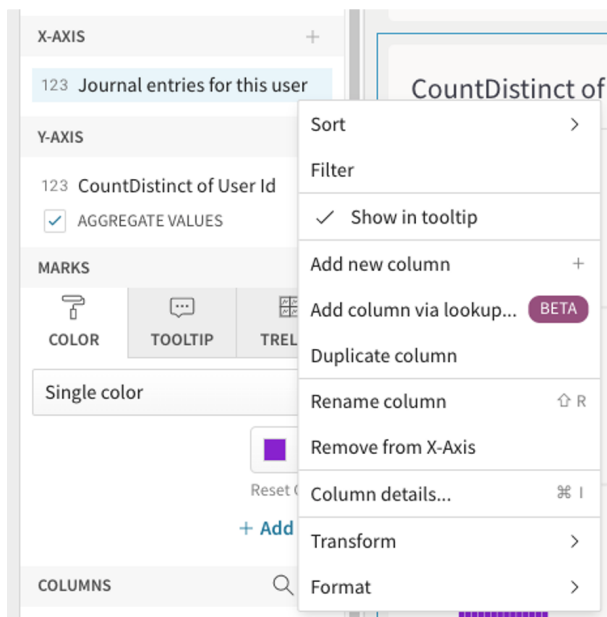


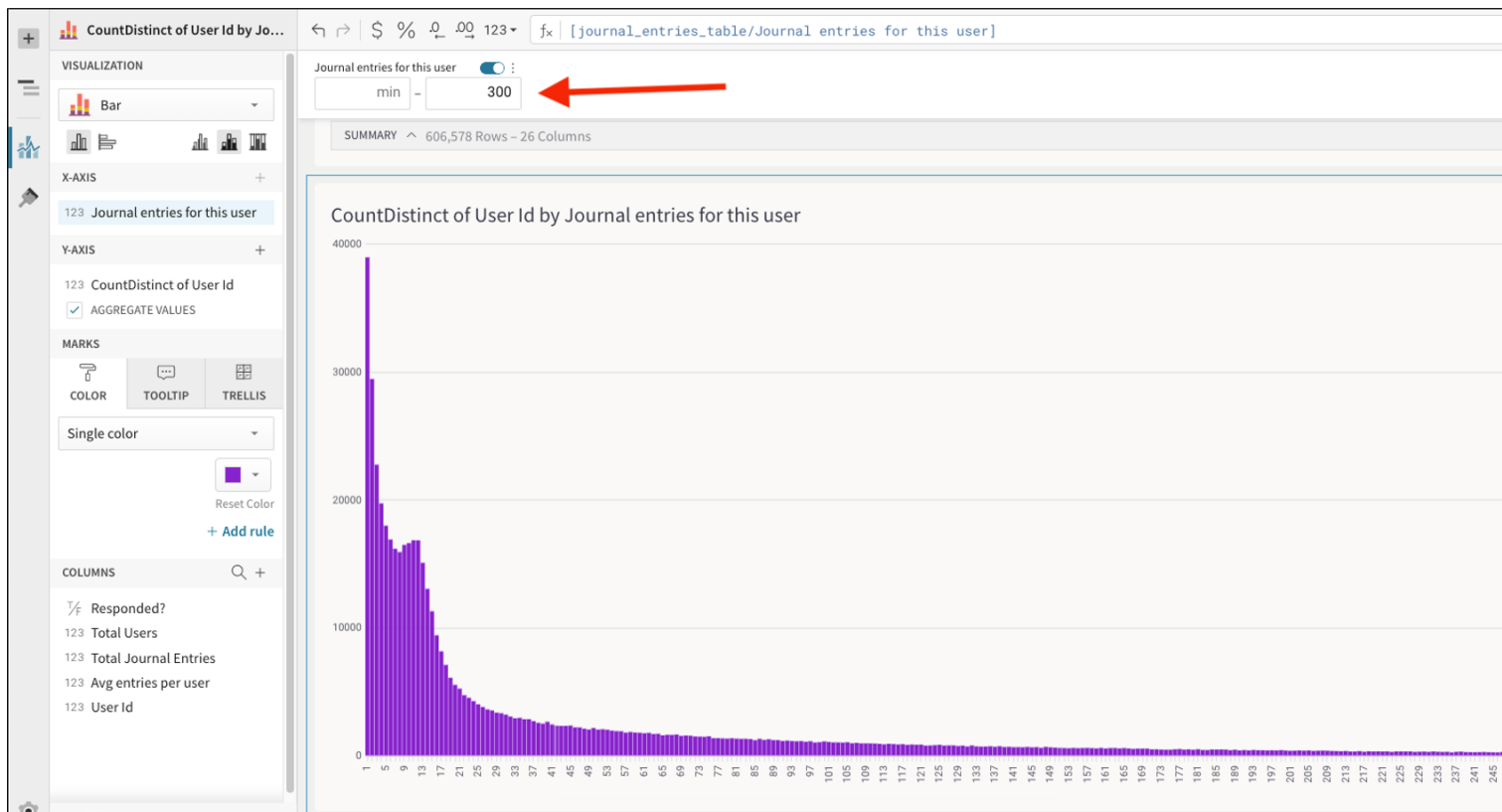
Notice that when we first drag "user ID" into the Y-axis, Sigma tries to automatically choose an aggregation for that variable. It's chosen sum, but we want countdistinct, so we select the dropdown and set aggregate → countdistinct. That gives us this chart:



This is getting closer — It is technically the distribution of journal entries per user, but there's a few ways we could make it more readable. We could filter out our "long tail" by limiting to only values less than, say, 300 entries. Or we could "bucket" the data into clumps of 10 entries. I'll show how to do both.

To filter on any of our variables, we just have to select the dropdown for that variable. So I'll filter on "journal entries for this user" and set the max to 300:

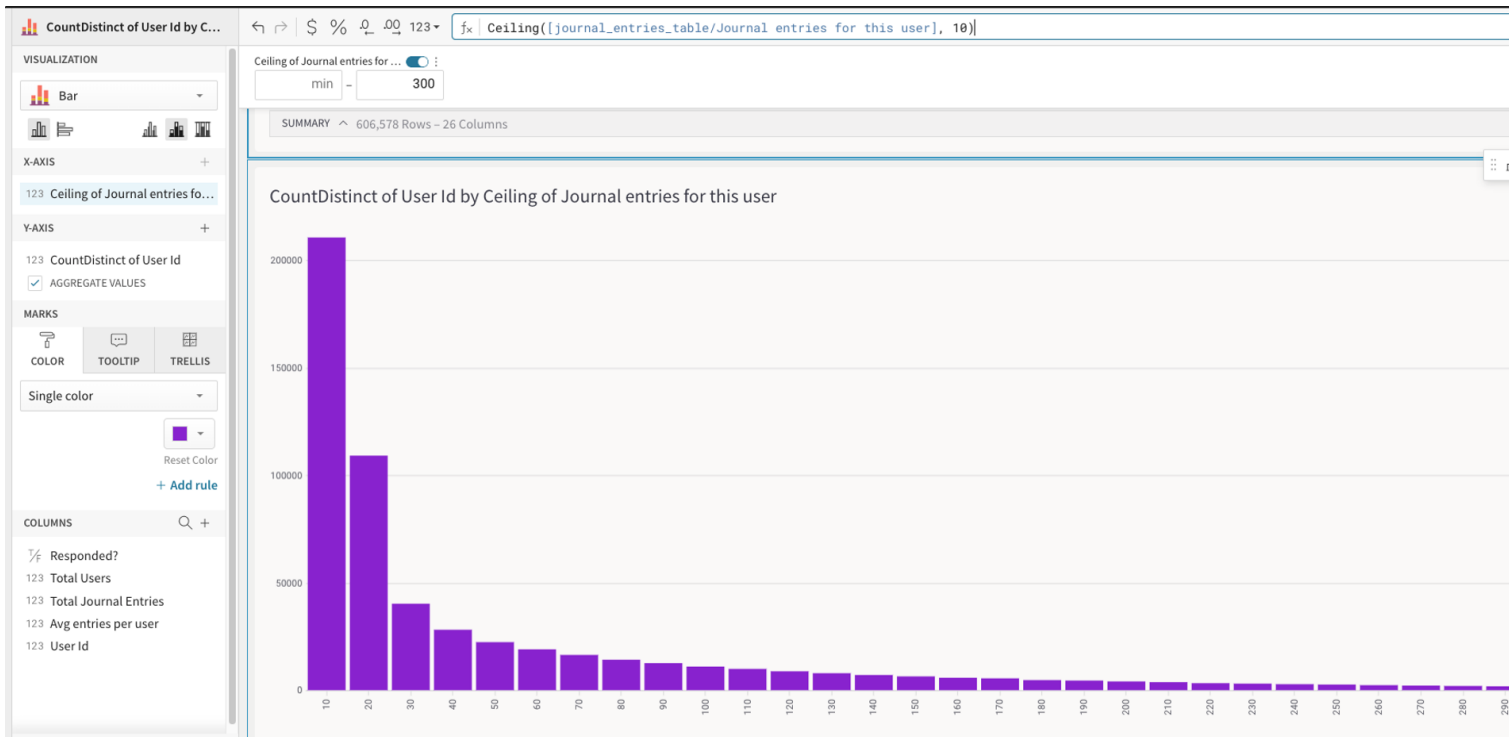




There, that looks a bit more manageable. We can see a slight bump in the curve around 13-14 entries, which makes sense if a lot of users stop after their free trial. But let's make it look even nicer. Instead of bucketing by each single increment up, let's bucket by 10s. For this, I'll use a formula called ceiling:

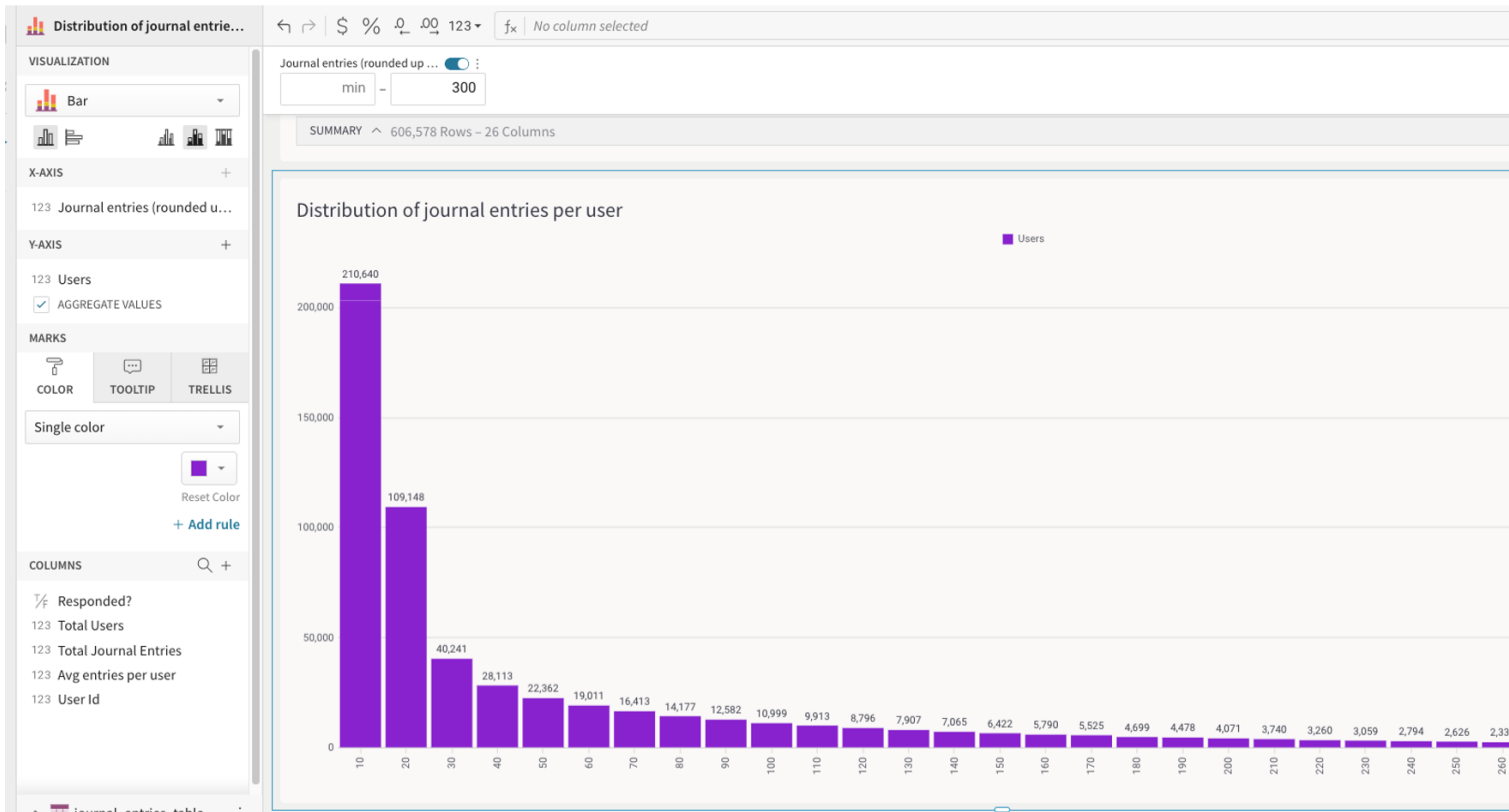
```
Ceiling([journal_entries_table/Journal entries for this user], 10)
```

This will set our Y-axis to the closest 10s value above the number of journal entries a user has. In other words, the "10" represents all users who have between 1-10 entries.



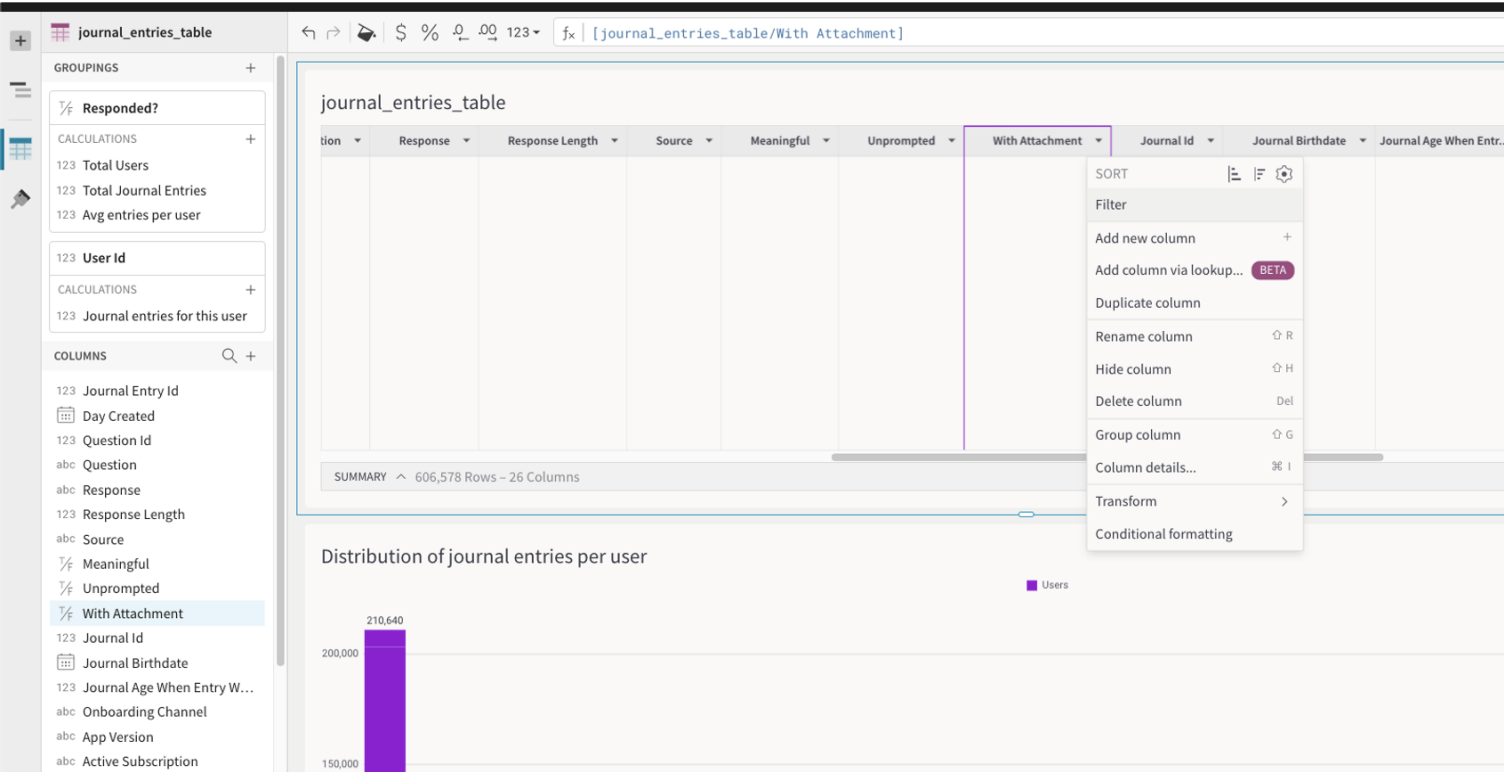
There we go, a nice steep engagement curve! (Although of course, we'd love for it to be flatter 😊)

Finally, we can title the chart and the axes whatever we want by double-clicking into their names, add a legend and data labels, and call it done!



What if we wanted to know the exact same thing, except for only photo entries? Do we have to rebuild this chart from an entirely new table? Am I asking leading questions to highlight the magic of dynamic filtering? I am!

If we want to filter this chart to only photo entries, all we have to do is filter the table it's based off of! I'll go back to the table and click on the dropdown for the "With Attachment" column (even though the data in it is hidden because we've rolled up to the user level, this still works!)



And there we go... We can see that the distribution of photo entries is even steeper.

